

Sarthak Singh

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EXPERIENCE

Influenced (Full Stack Developer)

July 2024 – June 2025

Java, Spring Boot, React JS, Node.js, AWS, DynamoDB

Remote, USA

- An AI-powered marketplace connecting brands with inventory to micro-influencers, using a generative AI recommendation engine to rank and match influencers with brand campaigns using NLP-based profile analysis for collaborative filtering.
- Developed an AI-powered marketplace connecting brands with micro-influencers, building RESTful APIs using Java and Spring Boot with MVC architecture, dependency injection, and JPA for database interactions.
- Implemented role-based access control and AI-driven workflows for product listings and collaboration management, with a React JS frontend consuming Spring Boot APIs backed by DynamoDB on AWS.

IBM (Software Development Engineer II)

July 2021 – Jan 2024

Python, AWS, GO, Java, Docker, Kubernetes, PostgreSQL, Nginx

Kochi, India

- Led the migration of from Elasticsearch to OpenSearch, redesigning indexing strategies and query optimization layers using Python and FastAPI to expose search APIs, reducing operational costs by up to \$3M annually and improving search query performance.
- Streamlined backend services in Golang for API management on IBM Cloud and AWS, leveraging Go's concurrency model with goroutines to handle high-throughput traffic routing via Nginx, reducing response times across distributed systems.
- Led the migration of secrets from SOS Vault to IBM Secrets Manager, optimizing the secrets retrieval caching layer to reduce latency while pulling the credentials, resulting in a more cost-effective secrets management infrastructure and earning me the IBM Cash Award.
- Containerized microservices with Docker, orchestrated deployments on Kubernetes (EKS), and automated CI/CD pipelines with Jenkins, improving deployment efficiency and system observability using CloudWatch and Grafana.

Informatica by Salesforce (Software Engineer Intern)

Jan 2021 – July 2021

Python, ETL, Oracle DB, Tableau, Java, AWS S3

Bangalore, India

- Contributed to customer support by resolving 100+ tickets, offering technical guidance and troubleshooting for Informatica ETL issues.
- Provided direct technical support for integrating the Informatica ETL tool with databases such as PostgreSQL, MySQL, and AWS S3.

PROJECT & RESEARCH

AI Powered Solutions

Oct 2025 – Dec 2025

LLM, RAG, Gradio, Streamlit, Groq, MCP, HF Pipelines, LangChain

[Link](#)

- RAG Voice Customer Support: An end-to-end voice-enabled customer support assistant combining Retrieval-Augmented Generation with real-time speech interaction.
- An AI-powered Indian Law Assistant leveraging RAG to ground responses in Indian legal statutes, utilizing MCP for seamless integration and processing of real-world legal data and case law analysis.

Methane Optimizer Tool

Aug 2024 – Dec 2024

React JS, JavaScript, MySQL, D3.js, Git

- Led the web development of a ReactJS application to visualize the optimization of methane production in a Urea plant, visualizing methane output levels by tweaking control parameters to optimized values generated by underlying CNN deep learning models.

Research Article (Journal of Environmental Management)

July 2022 – Jan 2024

[Link](#)

- Contributed to research on green procurement dynamics in the public sector, developing an assessment framework to help developing nations make data-driven decisions towards green environmental practices.

EDUCATION

University of Maryland

Master in Engineering, Software Engineering, Cumulative GPA: 3.6/4.0

Teaching Assistant | *Introduction to Software Engineering ENPM611*

College Park, MD

Jan 2024 – Dec 2025

Vellore Institute of Technology

Bachelor of Technology in Computer Science and Engineering

Vellore, India

June 2017 – May 2021

TECHNICAL SKILLS

Programming: Python, GO, FastAPI, Java, SQL (PostgreSQL, MySQL), NoSQL, Redis, Kafka, Django

DevOps & Cloud: Docker, K8, Jenkins, CI/CD, Terraform, AWS(CloudWatch, Grafana, EC2, EKS, IAM, S3, Lambda)

Systems: Microservices, Distributed Systems, Scaling Systems, Object-Oriented Design, Unit Testing

Other: Git/GitHub, Windows, Linux, Data Structures & Algorithm, Communication, Innovation

AI: Cursor, NLP(transformer models), LLM(Claude,GPT), Deep Learning ML (Pytorch, SGD, scikit learn)